



T: +44 (0) 1494 484 844 F: +44 (0) 1494 482 155

USA T: + 1 714 735 4413 (Based in the UK)

E: sales@aircraftmaterials.com



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17-4PH Stainless Steel Bar (AMS 5643/ AMS 5622)

Technical Data Sheet

17-4PH is a martensitic, precipitation-hardening stainless steel that provides an outstanding combination of high strength and hardness along with excellent corrosion resistance. 17-4PH has good fabricating characteristics and can be age hardened by a single step, low temperature treatment. 17-4PH is a versatile material widely used in the aerospace, chemical, petrochemical, food processing, paper and general metalworking industries.

Chemical Composition Limits

Weight%	C	Si	Mn	P + S max	Cr	Ni	Cb + Ta	Cu
AMS 5643	0.07	1.0 max	1.0 max	0.04 P 0.03 S	15.0 - 17.5	3.0 - 5.0	5 X C / 0.45	3.0 - 5.0
AISI 630	0.07	1.0 max	1.0 max	0.04P 0.03 S	15.0 - 17.5	3.0 - 5.0	0.15 - 0.45	3.0 - 5.0

Typical Mechanical Properties

Material	Condition	Ultimate Tensile Strength (ksi)	0.2 % Yield Strength (ksi)	Elongation % in 2D	% Reduction of Area	Rockwell C Hardness
Alloy 17-4PH Bar AMS 5643	Cond A	-	-	-	-	363 HB max
Alloy 17-4PH Bar AMS 5643	H900	190	170	10	40	388-444
Alloy 17-4PH Bar AMS 5643	H925	170	155	10	44	375-429
Alloy 17-4PH Bar AMS 5643	H1025	155	145	12	45	331-401
Alloy 17-4PH Bar AMS 5643	H1075	145	125	13	44	311-375
Alloy 17-4PH Bar AMS 5643	H1100	140	115	14	45	302-363
Alloy 17-4PH Bar AMS 5643	H1150	135	105	16	50	277-352

Material	Condition	Ultimate Tensile Strength (ksi)	0.2 % Yield Strength (ksi)	Elongation % in 2D (Values are for sheet <0.1874" thick)	% Reduction of Area	Rockwell C Hardness
Alloy 17-4PH Sheet/Plate AMS 5604	Cond A	-	-	-	-	38 max
Alloy 17-4PH Sheet/Plate AMS 5604	H900	190	170	5	-	40-47

Alloy 17-4PH Sheet/Plate AMS 5604	H925	170	155	5	-	38-45
Alloy 17-4PH Sheet/Plate AMS 5604	H1025	155	145	5	-	35-42
Alloy 17-4PH Sheet/Plate AMS 5604	H1075	145	125	5	-	33-39
Alloy 17-4PH Sheet/Plate AMS 5604	H1100	140	115	5	-	32-38
Alloy 17-4PH Sheet/Plate AMS 5604	H1150	135	105	8	-	28-37

Availability

17-4PH is available in [Bar \(Round, Flat, Hexagon, Square\)](#), Wire, Forgings, [Tube](#), Ring, [Plate, Sheet and Strip](#). View Stock List, [Plate Stock List](#) or contact [Sales](#) for further availability.

Specifications

[AMS 5604 Sheet, Strip and Plate](#)

AMS 5643 and 5622 Bars (Flat, Hex, Round, Square), Wire, Forgings, Tubing and Rings

ASTM A 693 Plate, Sheet and Strip

UNS : S17400

17-4 DBLE H1150

ASTM A 564 T 630 Cond H1150D

NACE MRO175/ISO 15156-3

17-4PH Cond H 1150

ASTM A 564

17-4PH Sol

AMS 5643

ASTM A 484

AMS 5604 Sheet, Strip Cold Rolled and Plate.

AMS 5622 (Vac Melt) Bars (Flat, Hex, Round, Square), Wire, Forgings, Tubing and Rings

AMS 5642 Bar

AMS 5643 Bars (Flat, Hex, Round, Square), Wire, Forgings, Tubing and Rings

MIL-S-25043

ASTM A564 Grade 630

ASTM A484 / ASME SA484

ASTM A564 / ASME SA564

ASTM A693 / ASME SA693 Plate, Sheet and Strip

WS 1.4542

X5CrNiCuNb16-4

AFNOR Z6CNU1704

UNS S17400

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"Just wanted to say thanks for your efforts and updates"

"I always try you as you often keep stock that can't be found elsewhere"

"Prompt answer & knowledge of available material, timescale & price within a good time"

"Packaging is superb"

"Thank you for your quick response"

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Disclaimer

Every effort is made to ensure that technical specifications are accurate. However, technical specifications included herein should be used as a guideline

only. All specifications are subject to change without notice.



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AircraftMaterialsUK.com Ltd Reg Co. in England and Wales 4141886 - Unit D, The Heights Business Park, Ibstone Road, Stokenchurch, Bucks, HP14 3BG.

Tel: + 44(0) 1494 484844 Fax: 44(0) 1494 482155 Tel US: +1 714 735 4413

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